Example 4.3

Find the differential equation of the family of curves $y = \frac{a}{x} + b$ where a and b are arbitrary constants.

Solution:

Given $y = \frac{a}{x} + b$

Differentiating w.r.t x, we get

$$\frac{dy}{dx} = \frac{-a}{x^2}$$

$$x^2 \frac{dy}{dx} = -a$$

Again differentiating w.r.t x we get

$$x^2 \frac{d^2y}{dx^2} + 2x\frac{dy}{dx} = 0$$

$$\Rightarrow x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} = 0$$ which is the required differential equation

Example 4.4

Find the differential equation corresponding to $y = ae^{4x} + be^{-x}$ where a, b are arbitrary constants,

Solution:

Given $y = ae^{4x} + be^{-x}$. (1)

Here a and b are arbitrary constants

From (1), $\frac{dy}{dx} = 4ae^{4x} - be^{-x}$ (2)

and $\frac{d^2y}{dx^2} = 16ae^{4x} + be^{-x}$ (3)

(1)+(2) $\Rightarrow$ $y + \frac{dy}{dx} = 5ae^{4x}$ (4)

(2)+(3) $\Rightarrow$ $\frac{dy}{dx} + \frac{d^2y}{dx^2} = 20ae^{4x}$

$$= 4\left(5ae^{4x}\right)$$

$$= 4\left(y + \frac{dy}{dx}\right)$$

$$\frac{dy}{dx} + \frac{d^2y}{dx^2} = 4y + 4\frac{dy}{dx}$$

$\Rightarrow$ $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} - 4y = 0$, which is the required differential equation.

Example 4.5

Find the differential equation of the family of curves $y = e^x(a\cos x + b\sin x)$ where a and b are arbitrary constants.

Solution :

$$y = e^x(a\cos x + b\sin x) \quad (1)$$

Differentiating (1) w.r.t x, we get

$$\frac{dy}{dx} = e^x(a\cos x + b\sin x) + e^x(-a\sin x + b\cos x)$$

$$= y + e^x(-a\sin x + b\cos x) \text{ (from (1))}$$

$$\Rightarrow \frac{dy}{dx} - y = e^x(-a\sin x + b\cos x) \quad (2)$$

Again differentiating, we get

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} =$$

$$e^x(-a\sin x + b\cos x) + e^x(-a\cos x - b\sin x)$$

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} =$$

$$e^x(-a\sin x + b\cos x) - e^x(a\cos x + b\sin x)$$

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} = \left(\frac{dy}{dx} - y\right) - y \text{ (from (1) and (2))}$$

$\Rightarrow$ $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2y = 0$, which is the requited differential equation.

Exercise 4.1

1. Find the order and degree of the following differential equations.

   (i) $\frac{dy}{dx} + 2y = x^3$

   (ii) $\frac{d^3y}{dx^3} + 3\left(\frac{dy}{dx}\right)^3 + 2\frac{dy}{dx} = 0$

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